# Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Python · AI in Production · Regulated Systems · Explainable AI · LLM Evaluation · Agentic Workflows · CI/CD · AWS · Database Design · Claude Code

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## PROFESSIONAL SUMMARY

Software Engineer with 8 years shipping production Python systems in regulated, compliance-critical environments - and two AI systems that demonstrate exactly what Lazarus AI is building: AI that moves from experimentation to production safely, with full auditability and explainability. AgentProbe is a CI-integrated LLM evaluation platform: a 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency) that gates every prompt against a regression suite before it reaches production - zero regressions shipped across 1,240+ test cases. LeaseI is an 8-agent LangGraph platform for regulated affordable housing compliance: every output is either cited to an exact source clause or flagged as unsupported; every eligibility and financial decision routes to human review - deterministic guardrails enforced in code, not just policy. At Entrata, owned the HUD income-limit data pipeline for 7 years: 180,000 domain records powering eligibility decisions for 100K+ housing units with 100% PHI compliance, zero security incidents, and rigorous database design across multi-tenant production schemas. Daily Claude Code user; Anthropic API integrated throughout. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## TECHNICAL SKILLS

**AI & Production Safety:** LLM Evaluation Frameworks, LLM-as-Judge, CI-Integrated Regression Gating, Auditable AI, Explainable AI, Hallucination Detection, HITL, Guardrails, Structured Outputs, Agentic Workflows, LangGraph, LangChain, Multi-Agent Orchestration, RAG Pipelines, ChromaDB, Vector Search, Embeddings, Prompt Engineering, Anthropic API (Claude), Claude Code, OpenAI API, Ollama, MCP, OCR, BERT, spaCy, NLP, PyTorch, scikit-learn

**Engineering & Backend:** Python (8yr), JavaScript, TypeScript, SQL, PHP, Bash, FastAPI, Flask, REST APIs, Microservices, Database Design & Management, PostgreSQL, MySQL, Oracle, NoSQL, Docker, CI/CD (GitHub Actions), Unit Testing, Debugging, AWS (S3, EC2, RDS, Glue), RabbitMQ, Redis, PySpark, Linux, Git

**Reliability & Regulated Systems:** Scalable & Maintainable System Design, Production Monitoring, Observability, Performance Engineering, Data Pipelines, ETL, PHI Compliance (Libsodium), Secure System Design, Code Review Standards, Data Integrity, NumPy, Pandas, Tableau, Agile/Scrum

## EXPERIENCE

### Magizhs Tech Solution

May 2025 - Aug 2025

**AI/ML Engineer Intern**

- Designed and deployed a production agentic AI system (LangGraph, FastAPI, Python) - multi-agent orchestration with RAG retrieval, deterministic guardrails, HITL escalation, and CI/CD-integrated eval gate enforcing quality at every release; leveraged Claude Code and AI-assisted development tools throughout the engineering lifecycle.
- Built observability on agent performance and retrieval latency; wrote unit tests and maintained regression suite ensuring auditable, maintainable outputs across all regulated decision paths.

### George Mason University

Aug 2024 - May 2026

**Graduate Teaching Assistant & Research Associate**

- Built Python ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through feature engineering and statistical validation across 190+ countries.
- Mentored 600+ students in Python, SQL, database design, and software engineering fundamentals; designed monitoring dashboards improving research comprehension by 40%.

### Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

**Senior Software Engineer · Lehi, UT (Remote)**

- **Scalable, maintainable production systems in a regulated domain** - 20+ Python and JavaScript backend modules at 99.5% uptime for 100K+ monthly active users; owned HUD income-limit database pipeline for 7 years (federal source → 180,000 domain records powering eligibility, rent, and tax-credit calculations); RabbitMQ event-driven async processing with retry, idempotency, and dead-letter guarantees; Docker containerization (50% scalability improvement); AWS S3, EC2, RDS, Glue.
- **Database design, security, and data integrity** - designed and optimized complex multi-tenant schemas for eligibility, financial, and document workflows; 200+ SQL/PostgreSQL/Oracle query optimizations (30% performance gain); 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents); Redis caching; AI-driven observability tooling reducing triage time 40%.
- **Code quality, debugging, and engineering excellence** - 300+ peer code reviews enforcing documentation-based coding standards; unit testing and CI/CD via GitHub Actions; maintained sub-10 bug count for 6+ consecutive months across multi-tenant production environments; mentored 5+ engineers on Python, database design, distributed systems, and maintainable code practices.

### Entrata India - Pune

Jan 2017 - Sep 2018

**Associate Software Engineer → Trainee**

- Built backend modules (PHP, Python, PostgreSQL) with database design for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

## PROJECTS

### AgentProbe: CI-Integrated LLM Evaluation & Production Safety Platform

Evals · CI/CD · Auditable AI

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation platform built to solve exactly the problem Lazarus AI addresses - AI moving safely from experimentation to production: 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, CI-integrated regression gate blocking any prompt that fails eval before it reaches production, multi-provider Anthropic/OpenAI/Ollama routing. Zero regressions shipped across 1,240+ test cases.

### Leasel: Auditable & Explainable AI for Regulated Compliance

Regulated · HITL · Explainable

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production AI system built for regulated affordable housing compliance: every output cited to an exact source clause or explicitly flagged as unsupported (100% citation correctness); deterministic guardrails on all eligibility and financial decisions; HITL escalation on low-confidence outputs - LLM extracts and reasons, deterministic or human logic decides every regulated call. 8-agent LangGraph pipeline with multi-source RAG; 93.6% precision, 87.9% F1 across 1,240 compliance test cases.

### AI on FHIR: NLP-Driven Medical Records Query System

Regulated · NLP · Full-Stack

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application translating natural-language queries into FHIR API calls via BERT embeddings and LLM inference - reduced retrieval time ~60%; HIPAA-compliant RBAC with authentication, authorization, and full data isolation across structured medical records.

## EDUCATION

**M.S. Data Analytics Engineering -** George Mason University, Virginia — May 2026 · GPA 3.87

**B.Tech Computer Science & Engineering -** JNTU Anantapur — 2013-2017